Course Overview

Python for AI-Driven Automation and Business Data Science

Course materials

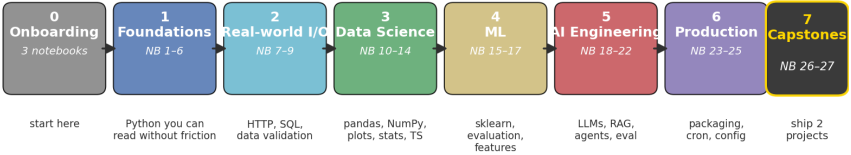
Contents

Outline

The course in one diagram

Python for AI-Driven Automation and Business Data Science

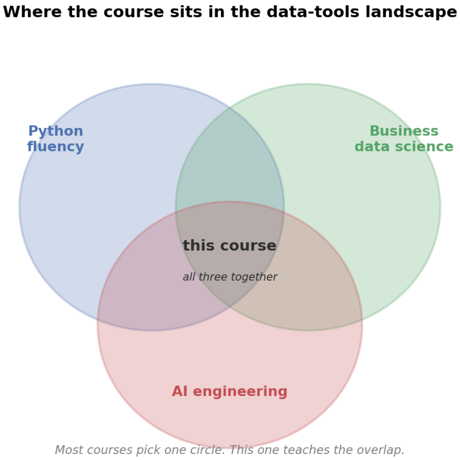
27 notebooks · 7 modules · 25–35 hours of focused study



—— runs entirely offline · every exercise has a worked solution · all 27 notebooks tested in CI ——

Foundations of every page: read prose → run code → tweak → predict → re-run

Where this course sits



Three skills, taught together

1. Python fluency

Read, write, and modify clean Python without friction. Functions, modules, error handling, defensive parsing.

2. Working with data professionally

pandas, NumPy, matplotlib, scikit-learn, statistics, time-series forecasting. Real datasets, not toy ones.

3. AI-driven automation

Calling LLMs as functions. Structured output. Retrieval-augmented generation. Tools and agents. Document processing. *Plus* the engineering wrapper around all of it.

Outline

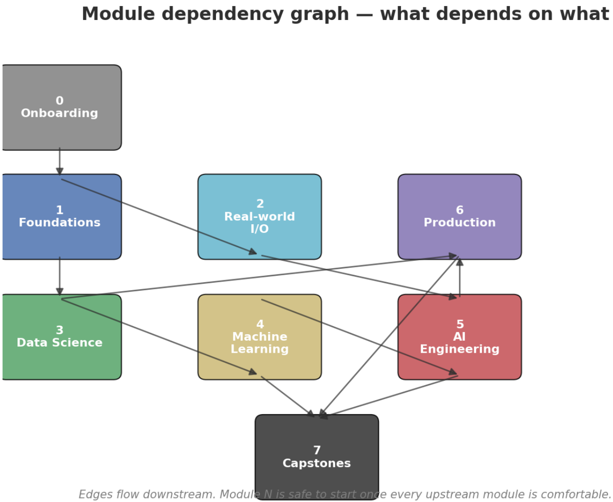
Seven modules, twenty-seven notebooks

#	Module	Notebooks
0	Onboarding	3 — start here, environment, study loop
1	Foundations	NB 1–6 — Python you can read without friction
2	Real-world I/O	NB 7–9 — HTTP, SQL, data validation
3	Data science	NB 10–14 — pandas, NumPy, plots, stats, time series
4	Machine learning	NB 15–17 — sklearn, model evaluation, features
5	AI engineering	NB 18–22 — LLM patterns, RAG, agents, evaluation
6	Production	NB 23–25 — packaging, scheduling, config
7	Capstones	NB 26–27 — analytics + AI assistant

Every module folder has its own README

Goals, prerequisites, what you'll build, the notebooks in order. Read it before you open the first notebook of the module.

Module dependencies — what comes before what

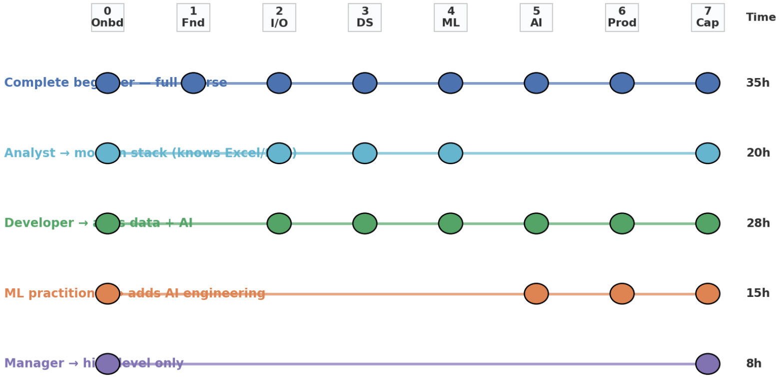


Edges flow downstream. Module N is safe to start once every upstream module is comfortable.

Outline

Pick the path that fits you

Pick the learning path that fits you



Each dot is one module. Lines connect the modules you'd touch on that path.

When to pick which path

Complete beginner

Never written code before. Take all seven modules in order. Time-budget: **35 hours**, split into 6 weeks of $\sim$6 h each.

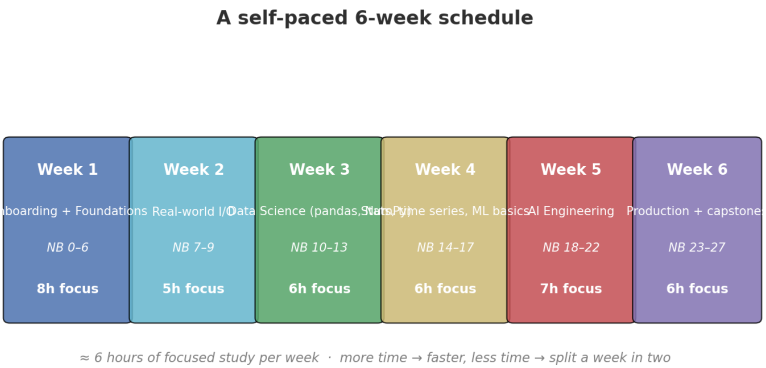
Analyst who knows Excel and SQL

Skim Module 1 (Python basics) and Modules 2 (SQL is already familiar). Focus on Modules 3 (DS), 4 (ML), 7 (capstone). Time-budget: **20 hours**.

ML practitioner adding AI engineering

Skim Module 4. Focus on Modules 5 (AI), 6 (Production), 7 (Capstone). Time-budget: **15 hours**.

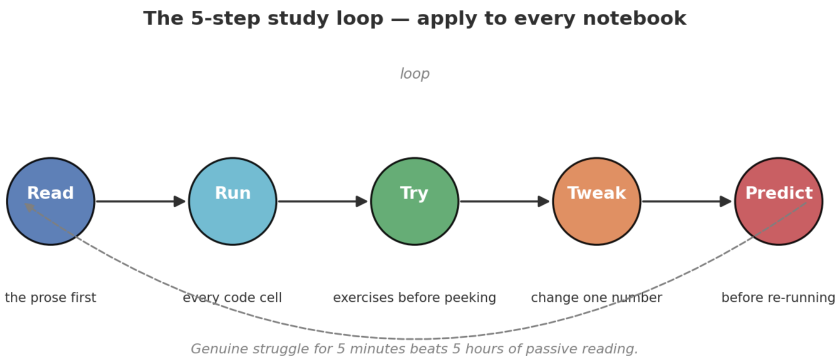
Six weeks at six hours each



Got more time? Move faster. Got less? Split one week into two.

Outline

The five-step study loop



Apply this loop to *every* notebook. Genuine struggle beats passive reading.

Six visual markers you'll see throughout

Marker	Meaning
`(Tip)`	A small "stop and notice" callout.
`(Intuition)`	The mental model behind the syntax.
`(Pitfall)`	A bug or anti-pattern that bites if you're not warned.
`(Exercises)`	Hands-on tasks. Try *before* peeking at the solution.
`(Bonus)`	One larger applied task per notebook.
`(Debug me)`	A cell that *intentionally* contains a bug for you to find.
`(Next step)`	Always points to the next notebook in your path.

Once you internalise the markers they become road signs, not decoration.

Outline

Six small artefacts you ship along the way

Six small artefacts you'll ship by the end

KPI snapshot
An AI-bot KPI report in ~30 lines of Python

Live API ETL
Daily weather pulled from Open-Meteo

SQL report
Channel summary written in SQL + pandas

Forecast
3-month forecast with Holt-Winters

Inbox triage
LLM that classifies + summarises feedback

Scheduled job
Weekly report task with retries + alerts

Plus two big capstones at the end

Capstone A — AI Support-Bot Analytics (NB 26)

Five channels × twelve months of operational data → 2×2 executive dashboard → regression demonstrating Simpson's paradox → five-bullet executive summary.

Capstone B — AI Customer-Feedback Assistant (NB 27)

Ingest free-form customer feedback → classify + extract via the MockLLM → ground answers in a knowledge base via RAG → orchestrate as a scheduled job with alerts. End-to-end AI feature.

Outline

What you need to install

For Google Colab

Nothing. All required libraries are pre-installed. Open the notebook, hit Shift+Enter.

For local Jupyter

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
jupyter lab
```

Notebook 0 includes an environment-check cell you run to confirm everything works before you move on.

Now → open Notebook 0

00_onboarding/00_master_onboarding

Read it. Run the environment check.

Then pick your path from the diagrams in this deck.

Welcome.

27 notebooks await you.